Shiyu Liu

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Education

Renmin University of China *Sep. 2024 – Present*
Dual-Degree Program in Finance and Data Science & Big Data Technology
School of Information, Jinke Experimental Class Beijing, China

Research Interests

Machine Learning • Large Language Models & LLM Systems • Algorithmic Game Theory & Mechanism Design • Online Decision-Making & Optimization
Current exploration: AI agents for cross-disciplinary scientific discovery and evidence-driven model revision.

Publications & Manuscripts

Peer-Reviewed Publications

[1] Yuqi Kong*, **Shiyu Liu***, Jiaxu Li, Hongtao Liu, Qi Qi, Weiran Shen. "**Modeling and Solving Stable Matching under Probabilistic Preferences with Large Language Models**." *ACL 2026*. [pdf]
* Equal contribution.

Manuscripts Under Review

[2] **Shiyu Liu**, Weiran Shen. "**SPARE: Selective Prefill Scheduling to Improve Throughput and Latency in LLM Serving**." *Under review at The Web Conference (WWW 2027)*. [pdf]

[3] Ren Liu, **Shiyu Liu**, Xiaodong Liu, Yuan Guo, Chen Yuanze, Lihua Zhang, Weiran Shen. "**Revenue Optimization in Repeated Auctions with Additive Boosts**." *Under review at The Web Conference (WWW 2027)*. [pdf]

Research Experience

Renmin University of China | Advisor: Prof. Weiran Shen

SPARE: Selective Prefill Scheduling to Improve Throughput and Latency in LLM Serving

First Author (Project Lead) Mar. 2026 – Present

- Investigated whether strategically delaying unfinished prefill computation could improve both first-token latency and latency-constrained serving capacity in memory-constrained LLM serving systems.
- Led the design and implementation of SPARE, a KV-cache-aware online scheduler that selectively defers unfinished prefills while protecting immediate output opportunities, using pressure-adaptive token budgets and chunk limits based on projected physical KV-block savings.
- Developed a GPU-calibrated, vLLM V1-aligned evaluation framework, achieving 99.24% feasibility agreement with GPU serving across four LLMs; led experimental evaluation and manuscript preparation.
- Demonstrated up to 2.65× serving capacity under matched mean-TTFT constraints and up to 82.1% lower mean TTFT at matched arrival rates, showing that strategic deferral can mitigate KV-cache contention and improve the latency–capacity trade-off.

Revenue Optimization in Repeated Auctions with Additive Boosts

Second Author Feb. 2025 – Jun. 2025

- Investigated revenue optimization in repeated second-price auctions, where intertemporal incentives can undermine truthful bidding despite incentive compatibility in individual auction rounds.
- Contributed to the design of a threat-based boost mechanism that uses future boost allocations to align bidders' incentives, and participated in theoretical analysis and proofs of Bayesian incentive compatibility.
- Led the implementation and experimental evaluation of the proposed mechanism, including revenue optimization using analytical gradients and projected gradient ascent, and conducted experiments with synthetic and real-world auction data.
- Demonstrated revenue improvements of at least 19.30% and 21.76% over the strongest static benchmarks for homogeneous and heterogeneous buyers, respectively; contributed to research discussions and manuscript preparation, highlighting how intertemporal incentives can improve revenue while preserving truthful bidding.

Modeling and Solving Stable Matching under Probabilistic Preferences with LLMs

Co-First Author Jun. 2025 – Jan. 2026

- Investigated stable matching under stochastic preferences, addressing the limitations of classical deterministic stability in capturing uncertain and boundedly rational decision-making.

- Designed an LLM-based probabilistic preference framework grounded in quantal response theory, introduced Expected Blocking Pairs (EBP) to quantify instability, and developed the Hybrid GS–LLM matching algorithm with probabilistic acceptance decisions.
- Led implementation, experimental design, and evaluation across five LLMs and 21 real-world speed-dating groups, analyzing LLM-generated preferences, switching behavior, and matching stability; authored the paper's Methodology section.
- Showed structured quantal-response patterns in LLM-induced decisions and lower expected instability than classical Gale–Shapley in most evaluated groups, demonstrating the value of uncertainty-aware matching mechanisms.

Cross-Disciplinary Scientific Discovery Agent

Independent Research (Ongoing) 2026 – Present

- Building an AI research agent to formulate testable questions across scientific disciplines by connecting domain knowledge and identifying gaps in existing explanations.
- Developing an evidence-driven model revision loop that challenges assumptions, proposes missing variables or alternative relationships, and evaluates revised hypotheses through counterexamples and targeted experiments.

Grants, Scholarships & Honors

2026 CCF Undergraduate Qihang Program (101 selected nationwide from computing undergraduates and first-year master's students).

2026 Beijing Municipal Natural Science Foundation – Undergraduate Qiyan Research Program Project Lead (Sole Student Investigator) (One of 200 selected from 582,387 undergraduates in Beijing (≈ 0.034%)).

2026 First Prize (Final Round), 17th JD Cup Student Innovation and Entrepreneurship Competition — CheeseX (74 entries).

2026 Third-Class Academic Scholarship, Renmin University of China.

2025 Outstanding Student Award (Sanhao Student), Renmin University of China.

2024 Meritorious Winner, Mathematical Contest in Modeling (MCM) (awarded to ≈6.3% of participating teams).

National Undergraduate Innovation Training Program Project Lead (reported selection rate: 4%).

National Undergraduate Entrepreneurship Training Program Project Lead (reported selection rate: 13.7%).

Third Prize, Chinese Mathematics Competition for College Students (Beijing Division) (municipal-level award).

Academic Service & Outreach

Student Representative & Speaker, CCF Qihang Program 2026

Delivered a speech on behalf of the undergraduate cohort at a CCF event, sharing research experiences and reflections on academic development.

Speaker, AI + Learning Academic Seminar Series 2026

Delivered student-oriented seminars on artificial intelligence, large language models, and practical AI tools, combining foundational concepts with hands-on demonstrations.

CheeseX / Zhishi Academic Platform

Core Team Member — Helped build an AI-driven platform for university project collaboration and early-stage research exchange.

Intelligent Legal AI Model, Renmin University of China Law School

Contributor — Participated in development of a domain-specific legal AI model.

Skills

AI-Native R&D	Advanced proficiency in OpenAI Codex CLI, Claude Code, and Cursor; AI-assisted codebase exploration, software development, debugging, and research experimentation.
Agentic Workflows	Model Context Protocol (MCP), tool/function calling, LLM API integration, agent orchestration, and multi-step AI workflows.
ML & LLM Systems	PyTorch, Hugging Face Transformers, vLLM; GPU-based benchmarking and experimental evaluation.
Programming	Python, C/C++, SQL; Linux, Git/GitHub, Docker, OpenMP.
Hobbies	Badminton, swimming, baseball, soccer, stand-up comedy, and board games .